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TISSUE CLOSURE DEVICE

Abstract

Device and method embodiments discussed herein are directed to mechanical closure of an access passage in a tissue layer adjacent to an access hole in a vessel such as an artery or vein of a patient. Some of these embodiments may also be applicable to direct closure of a vessel wall in some instances.

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a continuation of co-pending U.S. patent application Ser. No. 18/538,926, filed Dec. 13, 2023, by Henrik NYMAN et al. titled “TISSUE CLOSURE DEVICE”, which is a divisional of U.S. patent application Ser. No. 16/190,694, filed Nov. 14, 2018, by Henrik NYMAN et al. titled “TISSUE CLOSURE DEVICE”, which claims priority from U.S. Provisional Patent Application Ser. No. 62/587,353, filed Nov. 16, 2017, by Henrik NYMAN et al. titled “Device for Mechanical Approximation of Fascia”, each of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] In many percutaneous cardiovascular procedures, a catheter is inserted into an artery, such as the femoral artery, through a percutaneous vascular access. The catheter may be inserted, typically over a guidewire, directly into an artery (a “bareback” procedure), or the catheter may be inserted through a vascular introducer. When the procedure is complete, the physician removes the catheter and then removes the introducer from the vessel (if one was used). The physician then must prevent or limit the amount of blood that leaks through the vascular access. Physicians currently use a number of methods to close the vascular access, such as localized external compression, suture-mediated closure devices, plugs, gels, foams and similar materials.

[0003] However, such closure procedures may be time consuming, and may consume a significant portion of the time of the procedure. In addition, existing methods are associated with complications such as hematoma or thromboses. Still further, some of such procedures, particularly suture-mediated closure devices, are known to have high failure rates in the presence of common vascular disease such as atherosclerosis and calcification.

SUMMARY

[0004] Some embodiments of a vascular closure device may include a housing having an elongate configuration with an axial length greater than a transverse dimension thereof. Such a housing may further include a proximal end, a distal end and a distal section. A plurality of anchor deployers are slidably disposed within the housing adjacent each other at the distal section of the housing and are configured to extend and spread from the distal section of the housing. Each of the anchor deployers may include a deployment rod which is slidably disposed relative to the housing and which includes an elongate resilient configuration. Each deployment rod may also include a distal end that extends distally and radially from the distal section of the housing so as to spread out from other deployment rod distal ends. In some cases, the deployment rods may be configured to extend distally at the same time or simultaneously. Each anchor deployer may also include an anchor which is secured to the distal end of the deployment rod and which is configured to grip tissue such as the tissue of a fascia tissue layer. The vascular closure device embodiment may further include a tissue grip which is deployable from the distal end of the housing.

[0005] Some embodiments of a method for vascular closure may include disposing a distal end of the housing of the vascular closure device to a position adjacent the passage in the tissue layer and deploying a plurality of anchor deployers from a distal section of the housing. The anchor deployers may be so deployed by distally advancing deployment rods of the anchor deployers in a distal and radially outward direction from the housing into the tissue layer in positions disposed about the passage in the tissue layer. Respective anchors of the anchor deployers may then be secured to the tissue layer in positions disposed about the passage in the tissue layer. The deployment rods may then be proximally retracted back into the distal section of the housing so as to draw the anchors and respective tissue layer portions secured thereto together adjacent the distal section of the housing to gather the tissue and close the passage in the tissue layer. Thereafter, a tissue grip may be deployed over the anchors and onto the tissue layer portions gathered and secured to the anchors so as to secure the tissue layer portions together with the access hole closed or reduced. The anchors may then be released from the tissue layer portions which are secured together.

[0006] Some embodiments of a method for vascular closure may include disposing a distal end of the housing of the vascular closure device to a position adjacent the passage in the tissue layer and deploying a plurality of anchor deployers from a distal section of the housing. The anchor deployers may be so deployed by distally advancing deployment rods of the anchor deployers in a distal and radially outward direction from the housing into the tissue layer in positions disposed about the passage in the tissue layer. Respective anchors of the anchor deployers may then be secured to the tissue layer in positions disposed about the passage in the tissue layer. The deployment rods may then be proximally retracted back into the distal section of the housing so as to draw the anchors and respective tissue layer portions secured thereto together adjacent the distal section of the housing to gather the tissue and close the passage in the tissue layer. Thereafter, a tissue grip may be deployed over the anchors and onto the tissue layer portions gathered and secured to the anchors so as to secure the tissue layer portions together with the access hole closed or reduced. The anchors may then be detached from each of the respective deployment rods secured thereto and left in the patient secured to the tissue layer.

[0007] Some embodiments of a method for vascular closure may include disposing a vascular closure device adjacent a passage in a tissue layer which is disposed above and adjacent an access hole in a blood vessel of a patient and deploying a plurality of anchors from a distal section of the vascular closure device in a distal and radially outward direction therefrom. The method may also include engaging the tissue layer in positions disposed about the passage in the tissue layer with the anchors and securing the anchors to the tissue layer in the positions disposed about the passage in the tissue layer. Thereafter, the anchors may be proximally retracted closer together so as to draw the anchors and respective tissue layer portions secured thereto together thereby closing the passage in the tissue layer. A tissue grip may then be deployed onto the tissue layer portions drawn together by the anchors so as to secure the drawn together tissue layer portions thereby closing the passage in the tissue layer and achieving vascular closure of the access hole in the blood vessel.

[0008] Certain embodiments are described further in the following description, examples, claims and drawings. These features of embodiments will become more apparent from the following detailed description when taken in conjunction with the accompanying exemplary drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** schematically exemplifies a first embodiment of a vascular closure device according to a possible embodiment of the present disclosure.

[0010] FIGS. **2**A and **2**B show a detailed view of the creation of a tissue lock using the vascular closure device.

[0011] FIGS. **2**C and **2**D illustrate a closure sequence for treatment of an unwanted passage through a wall of a blood vessel without directly engaging the blood vessel.

[0012] FIGS. **3**A and **3**B conceptually illustrate an engagement member, exemplified as an anchor element.

[0013] FIGS. **4**A and **4**B illustrate the operation of an anvil member that functions as a deployable positioning feature.

[0014] FIG. **5** is an elevation view of a vascular closure device embodiment.

[0015] FIG. **6** is an enlarged view of the encircled portion **6-6** of the vascular closure device embodiment of FIG. **5**.

[0016] FIG. **6**A is a front view of the vascular closure device embodiment of FIG. **5**.

[0017] FIG. **7**A is an elevation view of a lock ring embodiment.

[0018] FIG. **7**B is a top view of the lock ring embodiment of FIG. **7**A.

[0019] FIG. **7**C shows a top view of the lock ring embodiment of FIG. **7**A in a radially expanded state.

[0020] FIG. **7**D is a transverse cross section of the lock ring embodiment of FIG. **7**B taken along lines **7**D-**7**D of FIG. **7**B.

[0021] FIG. **8**A is an elevation view of a lock ring embodiment.

[0022] FIG. **8**B is a top view of the lock ring embodiment of FIG. **8**A.

[0023] FIG. **8**C is a top view of the lock ring embodiment of FIG. **8**A in a radially expanded state.

[0024] FIG. **9** is an elevation view of a releasable anchor embodiment in a closed state.

[0025] FIG. **10** is an elevation view of the releasable anchor embodiment of FIG. **9** in an open state.

[0026] FIG. **10**A is a transverse cross section of the deployment rod of FIG. **10** taken along lines **10**A-**10**A of FIG. **10**.

[0027] FIG. **11** is an elevation view of a releasable anchor embodiment in a straightened state.

[0028] FIG. **12** is an elevation view of the releasable anchor embodiment of FIG. **11** in a curved, tissue gripping state.

[0029] FIG. **12**A is a transverse cross section of a deployment rod embodiment of FIG. **12** taken along lines **12**A-**12**A of FIG. **12**.

[0030] FIG. **13** is an elevation view of a detachable anchor embodiment.

[0031] FIG. **13**A is an enlarged view of the encircled portion **13**A-**13**A of the detachable anchor embodiment of FIG. **13**.

[0032] FIG. **13**B is a transverse cross section of the deployment rod embodiment of FIG. **13** taken along lines **13**B-**13**B of FIG. **13**.

[0033] FIG. **14** is an elevation view of a detachable anchor embodiment.

[0034] FIG. **14**A is a transverse cross section of the deployment rod embodiment of FIG. **14** taken along lines **14**A-**14**A of FIG. **14**.

[0035] FIG. **15** is an elevation view in partial section of a distal section of a housing of a vascular closure device disposed adjacent an access hole of a fascia tissue layer.

[0036] FIG. **16** shows the vascular closure device of FIG. **15** with the anchor deployers distally extended and with anchors thereof secured to the fascia tissue layer.

[0037] FIG. **16**A shows the releasable anchor embodiment of FIG. **9** in a closed state and secured to fascia tissue.

[0038] FIG. **16**B shows the releasable anchor embodiment of FIG. **11** in a curved tissue gripping state secured to fascia tissue.

[0039] FIG. **16**C shows the detachable anchor embodiment of FIG. **14** disposed in and secured to fascia tissue.

[0040] FIG. **16**D shows the detachable anchor embodiment of FIG. **13** disposed in and secured to fascia tissue.

[0041] FIG. **17** shows the anchor deployers of the vascular closure device of FIG. **16** being proximally retracted and closing the access passage in the fascia tissue layer.

[0042] FIG. **18** is an elevation view in partial section showing the lock ring being deployed from the distal end of the housing of the vascular closure device of FIG. **5**.

[0043] FIG. **19** shows the lock ring of FIG. **18** disposed around and securing portions of the fascia tissue layer gathered by the anchors of the vascular closure device of FIG. **5**.

[0044] FIG. **20** shows portions of the fascia tissue layer disposed about the access hole in a gathered and secured state with the lock ring and tissue adhesive disposed thereon.

[0045] FIG. **21** shows portions of the fascia tissue layer disposed about the access hole in a gathered and secured state with tissue adhesive only disposed thereon.

[0046] FIG. **22** shows portions of the fascia tissue layer disposed about the access hole in a gathered and secured state with the lock ring disposed thereon, the lock ring being further secured in fixed relation to the portions of fascia tissue by detachable anchors which are secured to the fascia tissue above the lock ring which may be mechanically captured by the deployed detachable tissue anchors.

[0047] FIG. **23** shows a lock ring embodiment in a relaxed self-constrained state disposed about and securing portions of gathered fascia tissue layer.

[0048] FIG. **24** shows upward and inward facing barb embodiments of the lock ring embodiment of FIG. **23** penetrating into gathered tissue of the portions of fascia tissue layer and mechanically preventing upward movement of the lock ring relative to the fascia tissue.

[0049] The drawings are intended to illustrate certain exemplary embodiments and are not limiting. For clarity and ease of illustration, the drawings may not be made to scale, and in some instances, various aspects may be shown exaggerated or enlarged to facilitate an understanding of particular embodiments.

DETAILED DESCRIPTION

[0050] After a minimally invasive vascular procedure, a hole in the form of an access passage or the like may be left in a major vessel at an access site that must be closed. Methods for percutaneous closure of such a hole may include remote suturing of the vessel, plugging the hole, and remote suturing of the fascia adjacent to the vessel. Certain device and method embodiments discussed herein are directed to mechanical closure of an access passage in the fascia tissue layer adjacent to an access hole in a vessel such as an artery or vein of a patient. Some of these embodiments may also be applicable to direct closure of an arterial wall in some instances. Some vascular closure device and method embodiments discussed herein may provide a robust and easy-to-use device for closing a vascular access hole after a minimally invasive procedure. In some cases, vascular closure device embodiments discussed herein may be useful for closing large vascular access holes. In addition, certain vascular closure device and method embodiments are discussed in U.S. patent application Ser. No. 15/277,542, filed Sep. 27, 2016, by Thomas Larzon, et al., entitled VASCULAR CLOSURE DEVICE, which is incorporated by reference in its entirety.

[0051] The following discussion of the device and method embodiments of FIGS. **1**-**4**B is directed generally to closure of a vascular access passage as well as axial positioning of certain portions of vascular closure device embodiments during such a closure procedure. Such axial positioning devices and methods may be applied to and used with any appropriate vascular closure device or method of embodiment discussed herein. Turning now to the drawings, and to FIG. **1** in particular, an embodiment of a vascular closure device **10** is introduced percutaneously over a guide wire **8** into a blood vessel/artery **5**, through the skin **1** and the fascia lata **2** of a patient. An optional anvil member **9** may be arranged inside the blood vessel **5** to create a reference point along an axial orientation to the engagement members **11** and/or for controlling bleeding from an inner lumen of the artery **5**. The engagement members **11** may then be placed and released through the vascular closure device **10** and may attach to fascia tissue **3** proximate to the blood vessel **5** and may involve the fascia membrane **3** (fascia iliacus), but, in some instances, not a wall **22** of the blood vessel **5**. The engagement members **11** may for example be pushed out of the vascular closure device **10** and into the fascia membrane **3** using deployment members provided as pusher rods **12** arranged in independent lumens provided with the vascular closure device **10**, for example through a pusher assembly in a common lumen that simultaneously deploys all engagement members **11**, through a spring-loaded mechanism or the like. For some embodiments, the engagement members **11** may be connected with a single filament such as a suture or a plurality of filaments or sutures **13**. In FIG. **1** there is further shown a femoral vein **4**, a femoral nerve **6** and adjacent/interstitial tissues **7**.

[0052] With further reference to FIGS. **2**A and **2**B, the suture **13** may for example be routed through each of the engagement members **11** in sequence. In particular, one suture **13** may be looped through each of the engagement members **11** in sequence, or a separate suture **13** may be attached to each engagement member **11**. The tissue, e.g. fascia membrane **3**, may then pulled together in a radially inward direction towards an access passage in the fascia layer **3** with the suture **13** connected to the engagement members **11**. When pulled together, the tissue/fascia membrane **3** is tightened towards the center and the access passage therethrough and may then create a tissue lock, thereby indirectly sealing the access hole **16** in the artery **5**.

[0053] That is, a distance between the initial position of the engagement members **11** and a distance between the engagement members once the engagement members **11** have been moved radially inward towards each other is thereby reduced. When tightening the fascia membrane **3** the anvil member **9** may be removed from the artery **5**.

[0054] Referring to FIGS. **2**C and **2**D, an embodiment of a vascular closure sequence is shown whereby a passage **16** through a wall **22** of the vessel **5** such as the blood vessel shown is treated such that leakage of blood from the interior volume of the blood vessel (not shown) is slowed or stopped to a clinically acceptable degree. As seen in FIG. **2**C, the passage **16** in the wall of the blood vessel, specifically, the femoral artery **5**, is disposed in general alignment with a passage **25** through the fascia tissue layer **3** disposed proximate to an outer surface of the femoral artery **5**. For this particular exemplary embodiment, the tissue layer disposed outside of and proximate to the outer surface of the femoral artery **5** is the fascia iliacus **3**. For purposes of this general discussion, the phrase “in general alignment” as applied to the respective passages **16**, **25** may mean at least that an appropriately sized elongate device such as a catheter or sheath may pass through both passages **16**, **25** without significant relative lateral displacement between the tissue **3** and artery **5**.

[0055] In addition, in some cases, the tissue layer **3** may be disposed sufficiently proximate the outside surface of the blood vessel **5** such that gathering and approximation of the fascia tissue **3** which is disposed about the passage **25** through the tissue **3** so as to close the passage **25** through the tissue/fascia membrane **3** and form a tissue lock is sufficient to tighten and displace the closed gathered tissue/fascia membrane **3** against the outer surface of the artery **5** which is adjacent the passage **16** through the artery **5** as shown in FIG. **2**D.

[0056] When the gathered tissue **3** has been displaced and deflected so as to be disposed against the passage **16** of the artery **5** and wall of the artery **5** disposed about the passage **16** in the artery **5**, this mechanical approximation will typically be sufficient in order to achieve a clinically sufficient slowing or stoppage of blood leakage from the passage **16** in the artery **5** in order to permit closure of an access site through the patient's skin **1** adjacent the passages. In some instances, an inner surface of the tissue layer **3** disposed proximate to the outer surface of the blood vessel **5** may be separated from the outer surface of the blood vessel in the region of the respective passages therethrough by a distance of up to about 10 mm, more specifically, up to about 5 mm.

[0057] With further reference to FIGS. **3**A and **3**B, there is conceptually illustrated an engagement member, exemplified as an anchor element **15**. In FIG. **3**A, the anchor element **15** is shown as initially deployed, so that it slides easily in the direction away from a deployment point. Note that the deployment point may optionally be deflected toward the tissue/fascia membrane **3** to promote engagement. FIG. **3**B shows the anchor element **15** after motion has been reversed toward the deployment point, and the anchor element **15** has embedded into the tissue/fascia membrane **3**. That is, a tip **17** of the anchors element **15** is in one embodiment hook-shaped, so that it easily slides outward without engaging the tissue/fascia membrane **3**. However, once the anchor element **15** is retracted, at least the tip **17** of the anchor element **15** is adapted to mechanically engage with the tissue/fascia membrane **3**.

[0058] FIGS. **4**A and **4**B conceptually illustrate the operation of an anvil member exemplified as a deployable positioning feature **20**. In FIG. **4**A, deployable positioning feature **20** may be inserted through the wall **22** and into the interior volume of the blood vessel, such as the femoral artery **5**. The deployable positioning feature **20** may be structured similar to an umbrella (using a mesh material), where the deployable positioning feature **20** in a radially collapsed form may be inserted into the artery **5**. Once within the artery **5**, with further reference to FIG. **4**B, the deployable positioning feature **20** may be “unfolded” and radially expanded from the collapsed form such that a total surface area proximate to the longitudinal axis of the deployable positioning feature **20** is increased and thus may be retracted towards the interior wall of the artery **5**. Accordingly, a reference point may be thereby established for further operation of the vascular closure device **10**.

[0059] For the vascular closure device embodiments **24** shown in FIGS. **5-24**, a mechanical “grabber” type device may be used to grab the fascia tissue layer **3** around an access hole **25** in the fascia (see FIG. **15**), pull it together, and apply a tissue grip type of device such as lock ring **27** or other tissue grip type retention mechanism such as a tissue adhesive **30** directly to the tissue of the fascia **3** to secure the gathered fascia tissue **32** disposed around the access hole **25** in the closed position to form a tissue lock and achieve vascular closure of an access hole **16** in the adjacent vessel **5**. FIGS. **5-24** illustrate embodiments of such grabber-type vascular closure devices **24**. The arms, also referred to herein as anchor deployers **26**, may be initially retracted into a housing **28** to achieve a low profile. Once placed above the fascia tissue layer **3**, the arms **26** may be extended distally and radially outward. Due to their preformed curved shape formed from an elastic resilient material (such as stainless steel or nitinol), the arms **26** may be configured to spread out into a radially dispersed pattern disposed around the access hole **25** in the fascia tissue layer **3**.

[0060] The arms **26** may then engage the fascia tissue layer **3** at two or more points around the access hole **25**. This engagement may be accomplished by an anchor **31** such as a small jaw **34** or the like mounted on the distal end **36** of a deployment rod **38** of each arm **26** and adapted to grab the fascia **3**, or a small hook **15**, **40** (see FIGS. **3**A and **12**) could be used to engage the fascia **3**. Other similar mechanisms may also be used. Once each arm **26** of the vascular closure device **24** has secured the fascia **3**, the housing **28** may be advanced down over the arms **26**, or the arms retracted proximally into the housing **28**, thereby pulling the distal ends **36** of the arms **26** together in a radially inward and proximal direction, approximating the edges of the fascia tissue layer **3** disposed about the access hole **25** therein, and closing or minimizing the access hole **25**.

[0061] In addition, the retention mechanism, such as a tissue grip mechanism that holds the approximated edges of the gathered fascia tissue **32** in this gathered configuration, such as the ring or clip **27**, may then be deployed onto the gathered tissue **32**. Thereafter, the arms **26** or anchors **31** disposed thereon may be disengaged from the fascia tissue layer **3** and the vascular closure device **24** withdrawn from the patient. Suitable tissue retention may be accomplished in some cases by use of the elastic resilient coil in the form of a lock ring **27** that is stretched onto the outside of the housing **28** about an outside circumference of the arms **26** or outside of a translation track of the arms **26**. The lock ring **27**, which may include an elastic and resilient self-contracting lock ring **27**, may be pushed off of the housing **28** by a distal end **42** of a larger, concentric outer tube **44** that lies more proximal over an inner tube **46** of the housing **28**. In some cases, the outer tube **44** may be actuated to advance distally relative to the inside tube **46** and deploy the lock ring **27** by advancing a lock ring actuator lever **47** which may be operatively coupled to the outer tube **44**. Once pushed off the inside tube **46** of the housing **28** and onto the gathered tissue **32**, such an elastic coil **27** may self-contract to circumferentially compress the gathered tissue **32** and brought together by the arms **26** in an inner radial direction and retain the gathered tissue **32** in a bunched or gathered configuration, thereby closing or reducing the access hole **25**.

[0062] Such elastic coil embodiments **27** may be made from high strength resilient materials such as stainless steel, nitinol, or the like. Some elastic coil embodiments **27** may also be made from or include bioresorbable and/or biodegradable materials. In some embodiments, the gathered tissue **32** may be held together by a biocompatible, rapidly curing tissue adhesive, such as cyanoacrylate, dispensed from a distal section **48** of the housing **28**. Such a tissue adhesive **30** may be dispensed as the gathered tissue of the fascia tissue layer **3** is coming together, to facilitate the apposition of surfaces containing tissue adhesive **30**, or the tissue adhesive **30** may be dispensed onto the gathered tissue **32** once the tissue is bunched together.

[0063] Referring to FIGS. **5** and **6**, some embodiments of the vascular closure device **24** may include the housing **28** having an elongate configuration with an axial length greater than a transverse dimension thereof. Such a housing **28** may further include a proximal end **50**, a distal end **52** and a distal section **48**. A plurality of anchor deployers **26** are slidably disposed within the housing **28** adjacent each other at the distal section **48** of the housing **28** and are configured to extend and spread from the distal section **48** of the housing **28**. Each of the anchor deployers **26** may include the deployment rod **38** which is slidably disposed relative to the housing **28** and which includes an elongate resilient configuration. Each deployment rod **38** may also include the distal end **36** that extends distally and radially outward from the distal section **48** of the housing **28** so as to spread out from other the distal ends **36** of the deployment rods **38**. In some cases, the deployment rods **38** may be configured to extend distally at the same time or simultaneously. Each anchor deployer **26** may also include the anchor **31** which is secured to the distal end **36** of the deployment rod **38** and which is configured to grip tissue such as the tissue of a fascia tissue layer **3**. The vascular closure device embodiment **24** may further include the lock ring **27** which may be deployable from the distal end **52** of the housing **28**. The vascular closure device of FIG. **5** includes 4 anchor deployers **26**, however any suitable number of anchor deployers **26** may be used. Some such vascular closure device embodiments **24** may include about 2 anchor deployers **26** to about 8 anchor deployers **26**, more specifically, about 3 anchor deployers **26** to about 5 anchor deployers **26**. For some embodiments, it may be desirable for the deployment rods **38** to have a rectangular transverse cross section profile to facilitate distal and radially outward extension while maintaining lateral stability. In some instances, the deployment rod embodiments **38** may have a generally flattened cross section profile with a radial transverse dimension that is less than a circumferentially oriented transverse dimension.

[0064] In some cases, the housing **28** may further include a guidewire lumen **54** extending an axial length thereof and a handle **56** secured to the proximal end **50** of the housing **28**. A deployment rod pusher **58** which may optionally be spring loaded with a resilient member such as a spring **60** in either a distally biased or proximally biased direction may be operatively coupled to the deployment rods **38** for actuation thereof. For deployment rod pusher embodiments **58** which are proximally biased this may correspond to a bias with the deployment rods **38** biased towards a retracted position which may be overcome by manual pressure in a distal direction against the deployment rod pusher **58**. The spring, such as spring **60** shown in FIG. **1**, may be used to provide such a resilient biasing force on the deployment rod pusher **58**. As shown, the deployment rod pusher **58** is operatively coupled to respective proximal ends of the deployment rods **38** and optionally configured to extend the deployment rods **38** simultaneously in a distal direction upon actuation.

[0065] In some instances, embodiments of a tissue grip mechanism may be disposed on the distal end **52** of the housing **28** around the anchor deployers **26** but generally not in contact with the anchor deployers **26** when in an undeployed state. The tissue grip mechanism may be configured to compress and secure gathered tissue portions **32** relative to each other in some cases. For such embodiments, once the tissue grip is deployed from the distal end **52** of the housing **28** over the anchor deployers **26** and respective anchors **31** thereof and onto the tissue which has been gathered and bunched by the proximally retracted anchor deployers **26**, different portions of the gathered tissue are secured in a fixed position relative to each other to form a tissue lock and, in some instances, vascular closure. In some cases, the tissue grip mechanism may include the lock ring **27** disposed about the anchor deployers **26** in the form of a self-retracting coil **27** with a central lumen **62** which may be sized to allow movement of the gathered tissue portions **32** disposed therein while the self-retracting coil **27** is in an expanded state as shown in FIGS. **7**C. The interior surface

64 of the central lumen **62** may be configured to compress and secure the gathered tissue portions **32** relative to each other when in a retracted compressed state as shown in FIGS. **7**A, **7**B, **20**, and **22**. In some cases, embodiments of the tissue grip may include the tissue adhesive **30** that may be dispensed from an outlet port, such as a distal port **66** of the guidewire lumen **54**, in the distal end **52** of the housing **28** as shown in FIGS. **20** and **21**. For some embodiments, such as the tissue adhesive **30** may include cyanoacrylate, fibrin glue, PEG-based polymers or the like.

[0066] In general, the tissue grip embodiments such as lock ring **27** and tissue adhesive **30** may optionally include a tissue engagement feature that is configured to stick to a surface of the gathered tissue **32** such as with the tissue adhesive **30** forming a bond with the gathered tissue **32** or sharp inner edge **64** of the diamond profiled lock ring **27** mechanically impinging the gathered tissue **32** so as to effectively secure the tissue portions together. Such tissue engagement features may further include features which may be mechanically captured within the gathered tissue **32** such as the barbs **76** of the "castle shaped" lock ring **27**′ shown in FIG. **8**A. The barbs **76** are configured to penetrate the gathered tissue **32** and be mechanically captured thereby as shown in FIG. **24**. All of these tissue engagement feature embodiments may generally be used to keep the tissue grip embodiments from slipping off of the gathered tissue **32** once deployed.

[0067] For some embodiments of vascular closure devices **24** discussed herein, the anchors **31** may include releasable anchors. For example, some releasable anchor embodiments may include the jaw **34** that can be moved between an open state and a closed state, each jaw **34** further including an opposed pair of tissue gripping teeth **68** as shown in the anchor embodiment **31** of FIGS. **9-10**A and **16**A. Such anchors **31** that include a jaw **34**, may be actuated by a pull wire **69** or the like that may apply tension to a hinged lever **71** at a proximal end of a hinged jaw portion **73** of the jaw **34** as shown in FIG. **10**. In some instances, the pull wire **69** or any other suitable actuation mechanism may be actuated by an actuation lever **59** which may optionally be disposed on the deployment rod pusher **58** shown in FIG. **5**.

[0068] For some releasable anchor embodiments **31**, each of the releasable anchors may have a shaft **70** with an elongate configuration having an axial length greater than an outer transverse dimension of the shaft **70**. The releasable anchor **31** may be configured to be actuated between a curved tissue gripping state having the curved distal end **40** as shown in FIG. **12** and a straightened state as shown in FIG. **11**. In some cases, the shaft **70** of the releasable anchor **31** may further include a sharpened distal end **72** which is configured to penetrate tissue such as the fascia tissue layer **3** when pushed distally, sharpened distal end **72** first, into the fascia tissue layer **3**. For some embodiments, the shaft **70** may be made from or include an optional shape memory metal that can be actuated between the curved tissue gripping state shown in FIG. **16**B and the straightened state shown in FIG. **11**. For such embodiments, the shape memory metal of the shaft **70** may include nickel titanium alloy. For other such embodiments, the shaft **70** may include an optional a bi-metal structure that can be actuated between the curved tissue gripping state and the straightened state.

[0069] In some cases, it may be desirable to use detachable anchors at the distal ends **36** of the deployment rods **38** and anchor deployers **26**. Referring to FIGS. **13-14**A, detachable anchor embodiments **31** are shown wherein the anchor deployers **26** include releasable junctions **74** disposed between each such detachable anchor **31** and respective deployment rod **38** which is secured thereto. In some cases, the detachable anchor embodiments **31** may include a proximally oriented barb **76**, and in some cases, a plurality of proximally oriented barbs **76**. For some embodiments, the releasable junctions **74** used to detach such detachable anchors **31** may include a releasable junction **74** released by thermal detachment, mechanical detachment or any other suitable detachment mechanism. The releasable junction embodiment **74** shown in FIG. **13**A includes a tether **78** which secured between the distal end **36** of the deployment rod **38** and anchor **31** and which may be melted by a heater element **80** disposed about the tether **78**. The heater element **80** may be actuated by applying electrical power from a power source **82** to the heater element **80** through a first electrically conducting wire **84** and a second electrically conducting wire

86 in electrical communication between the power source **82** and the heater element **80**.

[0070] Some embodiments of the vascular closure devices discussed herein may include a lateral surface configured to extend radially from a distal extension of the housing **28** while the lateral surface is disposed within a blood vessel **5** such as the anvil **9** shown in FIG. **1** and the deployable positioning feature **20** shown in FIGS. **4**A and **4**B and discussed above. Such a lateral surface **9**, **20** that extends radially may be used to provide a reference point between relative axial positions of the wall **22** of the blood vessel **5** and the anchors **31** prior to deployment of the anchors **31**. In some cases, embodiments of the anchors **31** may be deployed a predetermined axial distance from the wall **22** of the blood vessel **5** as measured from the lateral surface **9**, **20** which is disposed against the wall **22** of the blood vessel **5**.

[0071] Referring generally to FIGS. **15-24**, some embodiments of a method for vascular closure may include disposing the distal end **52** of the housing **28** of the vascular closure device **24** to a position adjacent the passage **25** in the tissue layer **3** and deploying a plurality of anchor deployers **26** from a distal section **48** of the housing **28**. The anchor deployers **26** may be so deployed by distally advancing deployment rods **38** of the anchor deployers **26** in a distal and radially outward direction from the housing **28** into the tissue layer **3** in positions disposed about the passage **25** in the tissue layer **3**. Respective anchors **31** of the anchor deployers **26** may then be secured to the tissue layer **3** in positions disposed about the passage **25** in the tissue layer **3** as shown in FIGS. **16**A-**16**B. The deployment rods **38** may then be proximally retracted back into the distal section **48** of the housing **28** so as to draw the anchors **31** and respective tissue layer portions secured thereto together adjacent the distal section **48** of the housing **28** to gather the tissue **3** and close the passage **25** in the tissue layer **3**. The deployment rods may be withdrawn with a spring or similar mechanism to apply a desired tension, or alternatively may be withdrawn with a lead screw, cam or similar mechanism to withdraw the deployment rods a desired distance independent of required tension. Thereafter, a tissue grip mechanism may be deployed over the anchors **31** and onto the tissue layer portions **32** gathered and secured to the anchors **31** so as to secure the tissue layer portions together with the access hole **25** closed or reduced. The anchors **31** may then be released from the tissue layer portions which are secured together as shown in FIGS. **20** and **21**. Thereafter, the deployment rods **38** and anchors **31** of each of the plurality of anchor deployers **26** may be proximally withdrawn into the distal section **48** of the housing **28**.

[0072] In some cases, for such method embodiments, it may be desirable after proximally retracting the anchors **31** and deployment rods **38** and gathering tissue of the tissue layer **3** and prior to deploying the tissue grip, to release the tissue layer **3** from the anchors **31** and re-deploy the anchor deployers **26** by re-extending the deployment rods **38** in a distal and radially outward direction and re-gripping the tissue layer **3** about the access passage **25** to reset the orientation and/or position of the anchors with respect to the access hole **25** before proximally retracting the anchors **31** and deployment rods **38** back into the housing **28**. This may be particularly useful when the operator is not satisfied with the initial placement of the anchors **31** about the access hole **25** in the tissue layer **3**. Such repositioning of the anchors **31** may be repeated as many times as is practical by the operator of the vascular closure device **24**.

[0073] In some instances, the anchors **31** may include the jaws **34** that can be moved between an open state and a closed state as shown in FIGS. **9-10**A. In such cases, securing the jaws **34** of the respective anchors **31** to the tissue layer **3** may include inserting tissue of the tissue layer **3** into the jaws **34** while the jaws **34** are in an open state and then closing the jaws **34** to the closed state to grip the tissue layer between the opposed surfaces of the jaws **34** as shown in FIG. **16**A. In some cases, the jaws **34** may include the tissue gripping teeth **68** on the opposed surfaces of the jaws **34** and gripping the tissue **3** between the opposed surfaces of the jaws **34** may include engaging and partially penetrating the tissue layer with the tissue gripping teeth **68**. In addition, for such embodiments, releasing the anchors **31** from the tissue layer portions may include moving the jaws **34** from the closed state to the open state such that the tissue is no longer gripped by the opposed surfaces of the jaws **34** or tissue gripping teeth **68** thereof.

[0074] For some embodiments of the vascular closure device **24**, the anchors **31** may include the shaft **70** which has the sharpened distal end **72** and which has an elongate configuration with an axial length greater than an outer transverse dimension as shown in FIG. **11**. The shaft **70** may be configured to be actuated between the curved tissue gripping state as shown in FIG. **12** and the straightened state as shown in FIG. **11**. For such embodiments, securing such respective anchors **31** to the tissue layer **3** may include inserting the sharpened distal ends **72** of the shaft **70** into the tissue layer **3** with the shaft **70** in a straightened state and actuating the shaft **70** to move the shaft **70** into a curved tissue gripping state about tissue of the fascia tissue layer **3** as shown in FIG. **16**B. For some such embodiments, releasing the anchors **31** from the tissue layer portions may include actuating the shafts **70** from the curved tissue gripping state to the straightened state and thereafter proximally retracting the shafts **70** from the tissue layer portions.

[0075] In some instances, deploying the tissue grip mechanism onto the tissue layer portions may include sliding the self-contracting ring **27** in an expanded state from the distal end **52** of the housing **28** and allowing the self-contracting ring **27** to contract to the relaxed state over the tissue layer portions **3** as shown in FIGS. **20** and **22**. The self-contracting ring **27** may optionally include a non-circular cross section, such as a diamond-shaped cross section, to increase friction with the tissue bundle **32**. Such a self-contracting ring **27** may also include a self-contracting ring embodiment **27**′ that includes barbs **76** which point upward and radially inward towards a longitudinal axis **88** of the self-contracting ring **27**′ as shown in FIGS. **8**A-**8***c*. Once deployed over tissue **32** and allowed to self-contract to a relaxed constrained state, as shown in FIGS. **8**A and **8**B, the barbs **76** may penetrate into the gathered tissue **32** and prevent the lock ring **27**′ from sliding off of the gathered tissue **32** as shown in FIGS. **23** and **24**. In some instances, deploying the tissue grip mechanism onto the gathered tissue **32** of the tissue layer portions **3** may include applying the tissue adhesive **30**, such as cyanoacrylate to the gathered tissue **32** of the tissue layer portions as shown in FIGS. **20** and **21**. FIG. **20** shows a tissue grip mechanism combination of the self-contracting ring **27** disposed over the gathered tissue **32** as well as the tissue adhesive **30** disposed thereon and in between different portions of the fascia tissue layer **3**.

[0076] Referring again to FIGS. **15-24**, some embodiments of a method for vascular closure may include disposing a distal end **52** of the housing **28** of the vascular closure device **24** to a position adjacent the access passage **25** in the tissue layer **3** and deploying a plurality of anchor deployers **26** from a distal section **52** of the housing **28**. The anchor deployers **26** may be so deployed by distally advancing deployment rods **38** of the anchor deployers **26** in a distal and radially outward direction from the housing **28** into the tissue layer in positions disposed about the passage **25** in the tissue layer **3**. Respective anchors **31** of the anchor deployers **26** may then be secured to the tissue layer **3** in positions disposed about the passage **25** in the tissue layer **3** as shown in FIGS. **16**C-**16**D. The deployment rods **38** may then be proximally retracted back into the distal section **48** of the housing **28** so as to draw the anchors **31** and respective tissue layer portions secured thereto together adjacent the distal section **48** of the housing **28** to gather the tissue **3** and close the passage **25** in the tissue layer **3**. Thereafter, the tissue grip embodiment may be deployed over the anchors **31** and onto the tissue layer portions **3** gathered and secured to the anchors **31** so as to secure the tissue layer portions together with the access hole **25** closed or reduced. The anchors **31** may then be detached from each of the respective deployment rods **38** secured thereto and left in the patient secured to the tissue layer **3**. In some cases, the anchor deployers **26** may include the releasable junctions **74** disposed between each respective deployment rod **38** and anchor **31** thereof. In such cases, detaching each anchor **31** from the respective deployment rod **38** secured thereto may include heating a tether **78** of the releasable junction **74** until it melts. In some instances, the anchor deployers **26** may be mechanically attached to the anchors **31** via a quarter-turn lock. In such cases, a cam or similar mechanism in the handle **56** may be used to rotate each of the deployment rods **38** a quarter turn or other suitable angular displacement to disengage the deployment rods **38** from

their respective anchors **31**.
[0077] Embodiments illustratively described herein suitably may be practiced in the absence of any element(s) not specifically disclosed herein. Thus, for example, in each instance herein any of the terms “comprising,” “consisting essentially of,” and “consisting of” may be replaced with either of the other two terms. The terms and expressions which have been employed are used as terms of description and not of limitation and use of such terms and expressions do not exclude any equivalents of the features shown and described or portions thereof, and various modifications are possible. The term “a” or “an” can refer to one of or a plurality of the elements it modifies (e.g., “a reagent” can mean one or more reagents) unless it is contextually clear either one of the elements or more than one of the elements is described. Thus, it should be understood that although embodiments have been specifically disclosed by representative embodiments and optional features, modification and variation of the concepts herein disclosed may be resorted to by those skilled in the art, and such modifications and variations are considered within the scope of this disclosure.
[0078] With regard to the above detailed description, like reference numerals used therein refer to like elements that may have the same or similar dimensions, materials and configurations. While particular forms of embodiments have been illustrated and described, it will be apparent that various modifications can be made without departing from the spirit and scope of the embodiments of the invention. Accordingly, it is not intended that the invention be limited by the forgoing detailed description.

Claims

1. A vascular closure device for closing a passage in a tissue layer located above and adjacent to an access hole in a wall of a blood vessel, the vascular closure device comprising: a housing having an elongate configuration with a proximal end and a distal section wherein the distal section is configured to be positioned above and adjacent to the passage in the tissue layer; a plurality of anchor deployers which are slidably disposed relative to the housing at the distal section of the housing and, each anchor deployer comprising: a deployment rod which is slidably disposed relative to the housing and which includes an elongate resilient configuration and a distal end that is configured to extend distally and radially outward from the distal section of the housing, and an anchor which is secured to the distal end of the deployment rod with a releasable junction including a detachment mechanism disposed between the detachable anchor and the deployment rod and which is directed distally and configured to engage and secure to a proximal facing of the tissue layer from above the tissue layer without passing through the passage in the tissue layer; and a tissue grip which is disposed on and deployable from the distal end of the housing and which is configured to be deployed directly to tissue of the tissue layer, compress and secure the tissue of the tissue layer, and maintain the tissue in a gathered configuration; wherein, when the tissue grip is deployed directly to the tissue of the tissue layer, the anchors are configured to be releasable from the proximal facing of the tissue layer, the deployment rods are configured to be retractable relative to the housing, and the tissue grip is configured to close the passage in the tissue layer and provide a tissue lock indirectly sealing the access hole in the blood vessel while the tissue layer is disposed above and adjacent the access hole in the wall of the blood vessel.
2. The vascular closure device of claim 1 wherein the housing further comprises a guidewire lumen extending an axial length thereof.
3. The vascular closure device of claim 1 further comprising a handle secured to the proximal end of the housing.
4. The vascular closure device of claim 1 further comprising a deployment rod pusher which is spring loaded and biased towards a retracted position and which is operatively coupled to the deployment rods.

5. The vascular closure device of claim 4 wherein the deployment rod pusher is operatively coupled to a proximal end of the deployment rods and configured to extend the deployment rods in a distal direction upon actuation.
6. The vascular closure device of claim 5 wherein the deployment rods are configured to extend and retract simultaneously.
7. The vascular closure device of claim 1 wherein the tissue grip comprises a lock ring disposed about the anchor deployers which includes a self-retracting coil with a central lumen which is sized to allow movement of the tissue portions while the self-retracting coil is in an expanded state and which has an interior surface of the central lumen that is configured to compress and secure the tissue portions relative to each other when in a retracted state.
8. The vascular closure device of claim 1 wherein the tissue grip comprises a tissue adhesive that may be dispensed from an outlet port in the distal end of the housing.
9. The vascular closure device of claim 1 wherein each of the releasable anchors comprises a jaw that can be moved between an open state and a closed state, each jaw further including an opposed pair of tissue gripping teeth.
10. The vascular closure device of claim 1 wherein each of the releasable anchors comprises a shaft having an elongate configuration with an axial length greater than an outer transverse dimension that is configured to be actuated between a curved tissue gripping state and a straightened state, the shaft further including a sharpened distal end.
11. The vascular closure device of claim 10 wherein the shaft comprises a shape memory metal that can be actuated between the curved tissue gripping state and the straightened state.
12. The vascular closure device of claim 11 wherein the shape memory metal of the shaft comprises nickel titanium alloy.
13. The vascular closure device of claim 10 wherein the shaft comprises a bi-metal structure that can be actuated between the curved tissue gripping state and the straightened state.
14. The vascular closure device of claim 1 wherein each of the detachable anchors comprises a proximally oriented barb.
15. The vascular closure device of claim 14 wherein the detachable anchors comprise a plurality of proximally oriented barbs.
16. The vascular closure device of claim 14 wherein the releasable junctions comprise a junction released by thermal detachment.
17. The vascular closure device of claim 14 wherein the releasable junctions comprise a junction released by mechanical detachment.
18. The vascular closure device of claim 1 further comprising a lateral surface configured to extend radially from a distal extension of the housing while the lateral surface is disposed within a blood vessel to provide a reference point between axial relative positions of the wall of the blood vessel and the anchors prior to deployment of the anchors.
19. The vascular closure device of claim 18 wherein the anchors can be deployed a predetermined axial distance from the wall of the blood vessel as measured from the lateral surface which is disposed against the wall of the blood vessel.
20. A vascular closure device, comprising: a housing having an elongate configuration with a proximal end and a distal section wherein the distal section is configured to be positioned above and adjacent to a passage in a fascia layer; a plurality of anchor deployers slidably disposed relative to the housing at the distal section of the housing, each anchor deployer comprising: a deployment rod slidably disposed relative to the housing, the deployment rod including an elongate resilient configuration and a distal end configured to extend distally and radially outward from the distal section of the housing, and an anchor secured to the distal end of the deployment rod with a releasable junction including a detachment mechanism disposed between the detachable anchor and the deployment rod, directed distally, and configured to engage and secure to a proximal facing of the fascia layer from above the fascia layer without passing completely through the passage in the

fascia layer; and a tissue grip deployable from the distal end of the housing and configured to be deployed directly to tissue of the fascia layer, compress and secure the tissue of the fascia layer, and maintain the tissue in a gathered configuration.